Maximum Likelihood for More Distributions

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Maximum likelihood for Poisson and exponential models

The score of a Poisson variable

Exercise 0.1 (Score function of a Poisson variable)**.** Derive the score function for a single Poisson random variable X, with respect to its mean parameter λ.

Solution

Solution 0.1. The log-likelihood of one Poisson observation is $\ell = x\log\{\lambda\} - \lambda - \log\{x!\}$, so:

$$
\begin{aligned}
\ell' &\overset{\text{def}}{=} \frac{\partial}{\partial\lambda}\ell && \text{(definition of the score)} \\
&= \frac{\partial}{\partial\lambda}(x\log\{\lambda\} - \lambda - \log\{x!\}) && \text{(Poisson log-likelihood)} \\
&= \frac{\partial}{\partial\lambda}(x\log\{\lambda\}) - \frac{\partial}{\partial\lambda}\lambda - \frac{\partial}{\partial\lambda}\log\{x!\} && \text{(linearity of differentiation)} \\
&= \frac{\partial}{\partial\lambda}(x\log\{\lambda\}) - \frac{\partial}{\partial\lambda}\lambda - 0 && \text{(derivative of a constant)} \\
&= \frac{\partial}{\partial\lambda}(x\log\{\lambda\}) - \frac{\partial}{\partial\lambda}\lambda && \text{(drop the zero term)} \\
&= x\frac{\partial}{\partial\lambda}\log\{\lambda\} - \frac{\partial}{\partial\lambda}\lambda && \text{(constant multiple rule)} \\
&= x\lambda^{-1} - \frac{\partial}{\partial\lambda}\lambda && \text{(derivative of } \log\lambda\text{)} \\
&= x\lambda^{-1} - 1 && \text{(derivative of } \lambda \text{ with respect to itself)} \\
&= \frac{x}{\lambda} - 1 && \text{(write the negative power as a fraction)} \\
&= \frac{x}{\lambda} - \frac{\lambda}{\lambda} && \text{(write both terms over the common denominator)} \\
&= \frac{x-\lambda}{\lambda} && \text{(combine the fractions)} \\
&= \frac{x - \mathrm{E}[X]}{\lambda} && (\mathrm{E}[X] = \lambda) \\
&= \frac{x - \mathrm{E}[X]}{\mathrm{Var}(X)} && (\mathrm{Var}(X) = \lambda)
\end{aligned}
$$

Theorem 0.1 (Score function of a Poisson variable)**.** *Let* $X \sim \text{Pois}(\lambda)$*, with mean* λ *and observed count* x*. The score* ℓ' *of the single observation* x *is the derivative of its log-likelihood* ℓ *with respect to* λ*. It equals the deviation of* x *from the mean* $\mathrm{E}[X]$*, divided by the variance* $\mathrm{Var}(X)$*:*

$$\ell' = \frac{x - \mathrm{E}[X]}{\mathrm{Var}(X)} = \frac{x-\lambda}{\lambda}$$

Proof

Proof. This is the solution to Exercise 0.1, worked in Solution 0.1. □

The score of an exponential variable

Exercise 0.2 (Score function of an exponential variable)**.** Derive the score function for a single exponential random variable X, with respect to the mean parameter μ.

Solution

Solution 0.2. The exponential density with mean μ is

$$\mathrm{p}(X = x) = \mu^{-1} e^{-x/\mu}$$

for $x > 0$, so $\ell = -\log\{\mu\} - \frac{x}{\mu}$, and:

$$\begin{aligned}
\ell' &\stackrel{\text{def}}{=} \frac{\partial}{\partial \mu}\ell && \text{(definition of the score)} \\
&= \frac{\partial}{\partial \mu}\left(-\log\{\mu\} - \frac{x}{\mu}\right) && \text{(exponential log-likelihood)} \\
&= \frac{\partial}{\partial \mu}(-\log\{\mu\} - x\mu^{-1}) && \text{(write } \frac{x}{\mu} \text{ as } x\mu^{-1}\text{)} \\
&= \frac{\partial}{\partial \mu}(-\log\{\mu\}) - \frac{\partial}{\partial \mu}(x\mu^{-1}) && \text{(linearity of differentiation)} \\
&= -\frac{\partial}{\partial \mu}\log\{\mu\} - x\frac{\partial}{\partial \mu}\mu^{-1} && \text{(constant multiple rule)} \\
&= -\mu^{-1} - x\frac{\partial}{\partial \mu}\mu^{-1} && \text{(derivative of } \log \mu\text{)} \\
&= -\mu^{-1} - x(-\mu^{-2}) && \text{(power rule)} \\
&= -\mu^{-1} + x\mu^{-2} && \text{(multiply)} \\
&= -\frac{1}{\mu} + \frac{x}{\mu^2} && \text{(write the negative powers as fractions)} \\
&= -\frac{\mu}{\mu^2} + \frac{x}{\mu^2} && \text{(write both terms over the common denominator)} \\
&= \frac{-\mu + x}{\mu^2} && \text{(combine the fractions)} \\
&= \frac{x - \mu}{\mu^2} && \text{(reorder the numerator)} \\
&= \frac{x - \mathrm{E}[X]}{\mu^2} && (\mathrm{E}[X] = \mu) \\
&= \frac{x - \mathrm{E}[X]}{\mathrm{Var}(X)} && (\mathrm{Var}(X) = \mu^2)
\end{aligned}$$

Theorem 0.2 (Score function of an exponential variable)**.** *Let X be an exponential random variable with mean μ and observed value $x > 0$. The score ℓ' of the single observation x is the derivative of its log-likelihood ℓ with respect to μ. It equals the deviation of x from the mean $\mathrm{E}[X]$, divided by the variance $\mathrm{Var}(X)$:*

$$\ell' = \frac{x - \mathrm{E}[X]}{\mathrm{Var}(X)} = \frac{x - \mu}{\mu^2}$$

Proof

Proof. This is the solution to Exercise 0.2, worked in Solution 0.2. □